



give me the fundamental explanations of shor's algorithm with the intention of explaining the need for quantum resistant technologies to setup a foundation for the research project

Shor's algorithm is a quantum algorithm capable of factoring large composite numbers exponentially faster than the best-known classical algorithms, directly threatening the security of widely used cryptographic systems like RSA and ECC that rely on the intractability of such factoring problems for their security. The advent and scaling of practical quantum computers would render these encryption schemes vulnerable, creating an urgent need to transition to quantum-resistant (post-quantum) cryptographic technologies. [\[1\]](#) [\[2\]](#) [\[3\]](#) [\[4\]](#) [\[5\]](#) [\[6\]](#)

Fundamentals of Shor's Algorithm

- Shor's algorithm works by transforming the problem of factoring a number into the problem of finding the period of a specific mathematical function, which it solves efficiently using a quantum Fourier transform—a task not feasible at scale for classical computers. [\[2\]](#) [\[7\]](#) [\[1\]](#)
- The core steps involve selecting a random integer, computing greatest common divisors (GCD), modular exponentiation, and—crucially—extracting the periodicity of the modular function via quantum means. [\[8\]](#) [\[1\]](#) [\[2\]](#)
- Whereas classical factoring algorithms are super-polynomial in running time, Shor's can factor large numbers in polynomial time (roughly $O((\log N)^3)$), making cryptographic keys breakable within a practically feasible time frame should large-scale quantum computers be available. [\[9\]](#) [\[7\]](#)

Why This Threatens Modern Cryptography

- The security of most public-key cryptosystems, especially RSA, ECC, and Diffie-Hellman, is predicated on the assumption that factoring large numbers or solving discrete logarithms is computationally unfeasible—an assumption invalidated by Shor's algorithm on quantum computers. [\[3\]](#) [\[10\]](#) [\[4\]](#) [\[11\]](#)
- If a cryptographically relevant quantum computer (CRQC) is built with enough qubits and reliability, it could break current encryption schemes, making private keys and sensitive data recoverable to adversaries. [\[12\]](#) [\[6\]](#) [\[3\]](#)
- The threat is not just theoretical or far off; an adversary could intercept and store encrypted traffic today and decrypt it in the future when quantum technologies mature. [\[13\]](#) [\[6\]](#) [\[3\]](#)

The Urgency for Quantum-Resistant Technologies

- Quantum-resistant or post-quantum cryptography is designed to remain secure in the face of quantum attacks, using mathematical problems believed to be hard even for quantum computers (like lattice-based and hash-based problems).^{[14] [15] [16] [13]}
- Major international bodies, such as NIST, are accelerating the standardization and implementation of quantum-resistant algorithms, with sector-wide transitions recommended well before practical quantum computers come online.^{[17] [16] [14]}
- Proactive adoption of these algorithms now is critical; waiting until quantum computers are a reality risks exposure of today's encrypted secrets and future data.^{[6] [3] [17] [13]}

In summary, Shor's algorithm demonstrates that the mathematical foundation of today's most prevalent cryptographic protocols is at risk from quantum advances, driving the urgent global push toward quantum-resistant cryptographic research and deployment.^{[3] [13] [6]}

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1. https://en.wikipedia.org/wiki/Shor's_algorithm
2. <https://www.quera.com/glossary/shors-algorithm>
3. <https://postquantum.com/post-quantum/shors-algorithm-a-quantum-threat/>
4. <https://www.fortinet.com/resources/cyberglossary/shors-grovers-algorithms>
5. <https://www.fortinet.com/uk/resources/cyberglossary/shors-grovers-algorithms>
6. <https://www.ssh.com/academy/how-quantum-computing-threats-impact-cryptography-and-cybersecurity>
7. <https://quantum.cloud.ibm.com/docs/tutorials/shors-algorithm>
8. <https://www.classiq.io/insights/shors-algorithm-explained>
9. <https://www.qutube.nl/quantum-algorithms/shors-algorithm>
10. <https://secureitconsult.com/quantum-computing-threatens-encryption/>
11. <https://www.lfdecentralizedtrust.org/quantum-threat-of-blockchain-and-cryptographic-systems>
12. <https://www.spinquanta.com/news-detail/shors-algorithm>
13. <https://quside.com/quantum-resistant-cryptography/>
14. <https://www.computer.org/publications/tech-news/trends/quantum-resistant-cryptography>
15. <https://www.ncsc.gov.uk/whitepaper/next-steps-preparing-for-post-quantum-cryptography>
16. <https://www.nist.gov/news-events/news/2022/07/nist-announces-first-four-quantum-resistant-cryptographic-algorithms>
17. <https://bpi.com/quantum-computing-the-urgent-need-to-transition-to-quantum-resistant-cryptography/>
18. <https://www.youtube.com/watch?v=FRZQ-efABeQ>
19. <https://algassert.com/post/1718>
20. https://en.wikipedia.org/wiki/Post-quantum_cryptography